Purified Contrastive Learning With Global and Local Representation for Hyperspectral Image Classification

Lin Zhao, *Member, IEEE*, Jia Li, Wenqiang Luo, Er Ouyang, Jianhui Wu, Guoyun Zhang, *Member, IEEE*, and Wujin Li

***Abstract*— Contrastive learning has emerged as a promising technique for hyperspectral image (HSI) classification. However, the inherent limitation of sliding window sampling in HSI results in partial samples within a mini-batch exhibiting extremely high similarity. Consequently, there is an increased number of negative sample pairs composed of similar samples, significantly reducing the effectiveness of contrastive learning. Moreover, prevailing classification models heavily depends on convolutional operations, emphasizing the extraction of local features but struggle to capture long-distance dependencies in both spatial and spectral dimensions. To address these problems and fully leverage the abundance of unlabeled samples, we propose a novel purified contrastive learning (PCL) framework for HSI classification. We design a complementary spatial–spectral representation encoder architecture that combines convolutional neural network (CNN) and Transformer to capture local features and global dependencies. More importantly, a purified contrastive loss function is proposed based on super-pixel spatial prior. Extensive experiments on three public datasets demonstrate the superiority of PCL over state-of-the-art methods in HSI classification. The code for this work is available at https://github.com/zhaolin6/PCL for the sake of reproducibility.**

***Index Terms*— Contrastive learning, convolutional neural network (CNN), hyperspectral image (HSI) classification, vision Transformer.**

I. Introduction

HYPERSPECTRAL remote sensing technology plays a key role in Earth observation research by providing a wealth of spatial and spectral data. Hyperspectral image (HSI) provides detailed spectral information over different land covers, allowing for the extraction of continuous spectral curves from different image elements. Due to this unique advantage, HSIs find a wide range of applications in the tasks, such as land-cover classification, spectral unmixing, object detection, agricultural monitoring, and marine environment monitoring [1], [2], [3]. Among these applications, HSI classification is one of the most critical steps, which involves the accurate assignment of semantic labels to each image element within the HSI [4].

In the literature on HSI classification, extensive research has been conducted in the literature on HSI classification, predominantly focusing on supervised learning [5], [6]. Over the past decade, it has seen an enormous success in deep learning approaches due to their powerful feature extraction capability, which can automatically derive valuable features from the HSI. The 1-D CNN was first introduced to effectively extract spectral features in land-cover classification tasks [7], [8]. Subsequently, various efforts have been conducted to obtain discriminative spatial–spectral features for HSI classification [9], [10], [11], [12], [13], [14], [15]. Notably, Paoletti et al. [9] proposed a deep 3-D convolutional neural network (CNN) architecture for HSI classification. Hang et al. [10] leveraged attention mechanism to enhance the CNN network, which further enhanced its feature extraction ability for HSI. Zhong et al. [13] designed a spectral–spatial residual network (SSRN) to extract spatial and spectral features from HSI. Hong et al. [14] proposed a Transformer-based architecture for more efficient extraction of spectral information from neighboring bands in HSI. Gao et al. [15] utilized a hierarchical stacking module to fuse shallow and deep information, extracting more discriminative features. Furthermore, recent advancements have led to the proposal of Mamba-based HSI classification methods, aiming to improve computational efficiency while achieving capabilities of Transformers for large-scale observations in remote sensing scenarios [16], [17].

The deep learning-based methods mentioned above have demonstrated impressive classification performance. However, training a deep learning model with a large number of parameters requires sufficient labeled samples [18]. Unfortunately, manually annotating semantic labels for large-scale datasets is not only a time-consuming and costly endeavor, but also it is prone to mislabeling. Consequently, ensuring an adequate number of samples for supervised HSI classification poses a challenge. In recent years, there has been increasing attention given to the extraction of valuable and complementary features from heterogeneous data [19], [20], [21], [22]. Moreover, these methods exclusively utilize labeled samples as the training set,

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Lin Zhao, Jia Li, Wenqiang Luo, Jianhui Wu, Guoyun Zhang, and Wujin Li are with Hunan Engineering Technology Research Center for 3D Reconstruction and Intelligent Application, Hunan Institute of Science and Technology, Yueyang 414000, China (e-mail: gyzhang@hnist.edu.cn; liwj@hnist.edu.cn).

Er Ouyang is with Shenzhen University, Shenzhen 518060, China.

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thereby ignoring the abundant and easily accessible unlabeled samples. This results in significant waste for sample-driven deep learning methods.

Recently, there has been encouraging progress in unsupervised visual representation learning, particularly using the methods related to the contrastive loss [23], [24], [25], [26]. Instead of expensive annotations, contrastive learning methods learn abstract semantic representation from unlabeled samples. The fundamental idea is to bring the similar samples close to each other while push away the dissimilar samples [26]. Some advanced classification methods based on contrastive learning have been proposed, surpassing supervised learning methods with extremely limited labeled samples [27]. In [28], the positive pairs consist of the images with adjacent spatial locations, and contrastive learning paradigm is employed to learn the smooth representations. In addition, to obtain higher quality representations, the DMVL [29] and L2CSL [30] have made an in-depth study on the data augmentation methods of HSI. Zhu et al. [29] improved the performance of contrastive learning by constructing image blocks of different scales and improving the CNN models accordingly. Hang et al. [12] leveraged the relationship between HSI and LiDAR data to extract features and transferred these features to enhance the HSI classifier.

Despite the promising performance demonstrated by the aforementioned contrastive learning methods, the progress in HSI encounters a bottleneck due to the specific patch-based sampling pattern. As illustrated in Fig. 1, spatially adjacent patches exhibit substantial overlap across large areas, resulting in remarkably high similarity. Consequently, this leads to an increased number of negative sample pairs with exceedingly high similarity, referred to as "false negative sample pairs." These false negative sample pairs pose a challenge to the contrastive learning paradigm, which aims to learn representations by bringing positive samples closer while simultaneously pushing negatives farther apart. In computer vision, limited attention has been given to the issue of false negatives. Huynh et al. [31] tackled the issue by identifying false negatives and implementing false negative cancellation strategies. However, it is important to note that the strategies tailored for natural images may not be suitable for HSI patches due to the inherent limitations of sliding window sampling in HSI.

To elaborate further, in Fig. 1, x_1 represents an anchor, and the remaining samples from the same batch are treated as negative samples. Among these negative samples, instances, such as x_2 and x_3, may naturally share similarity with x_1. Ideally, these samples should be brought closer to the x_1, but instead, they are actually being pushed away. This deviation from the principle of instance discrimination significantly impacts the overall performance of contrastive learning in the HSI domain.

In contrastive learning, the encoder severs as a pivotal component that provides effective representations for downstream tasks, such as classification. Broadly, the structure of the encoder can be mainly categorized into CNN and Transformer. CNN has demonstrated effectiveness in local spatial features extraction through the strategic stacking convolutional kernels, making them dominant in the field of HSI classification [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]. More recently, Transformer has been incorporated to HSI classification due to their ability to capture long-range dependencies [42], [43]. Despite various innovations and advancements introduced by the encoder in contrastive learning, it typically adopts a shared single-branch structure. However, this single-branch encoder cannot utilize CNN and Transformer simultaneously, leading to limitations in the encoder's ability to fully exploit the unique strengths inherent of both CNN and Transformer architectures.

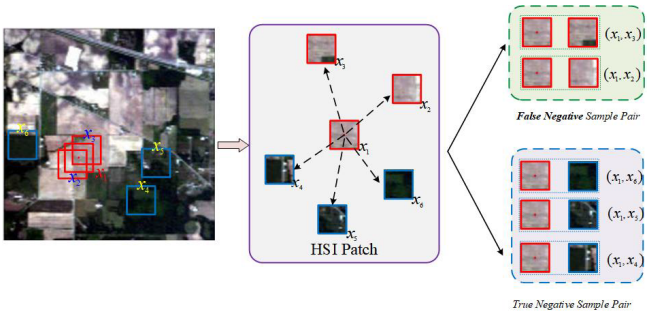

Fig. 1. Schematic of the false negative sample pair phenomenon.

False negative samples and encoder structure are fundamental problems in contrastive learning for HSI classification. In this article, we investigate these issues and propose a novel framework called purified contrastive learning (PCL) to mitigate the effect of false negative samples on contrastive learning and enhance the semantic representations of encoder. To effectively extract the spatial and spectral features of HSIs in the contrastive learning, we tailored a complementary spatial–spectral representation encoder and adopt a multiview training strategy. The contributions of this article are summarized as follows.

1) We propose a novel PCL framework with a modified network and a unique loss function termed "purified loss." Notably, the purified loss can mitigate the influence of negative sample conflicts. To the best of our knowledge, this is the first attempt to address the challenge of false negative samples in contrastive learning for HSI classification.
2) We design a complementary representation encoder that combines the strengths of both CNN and Transformer. It comprises two subnetworks: the spatial–spectral vision Transformer (SSViT) and the deep CNN (DCNN). The proposed architecture diverges from the commonly adopted shared single-branch encoder structure prevalent in the RGB domain.
3) Extensive experiments on benchmark datasets demonstrate that the PCL framework effectively alleviates the negative impact of the false negative samples on the contrastive learning and outperforms other state-of-the-art HSI classification methods.

II. Methodology

The overall architecture of PCL is depicted in Fig. 2, comprising three main components: Part A represents the complementary representation encoder, Part B involves the purified contrastive loss based on superpixel segmentation, and

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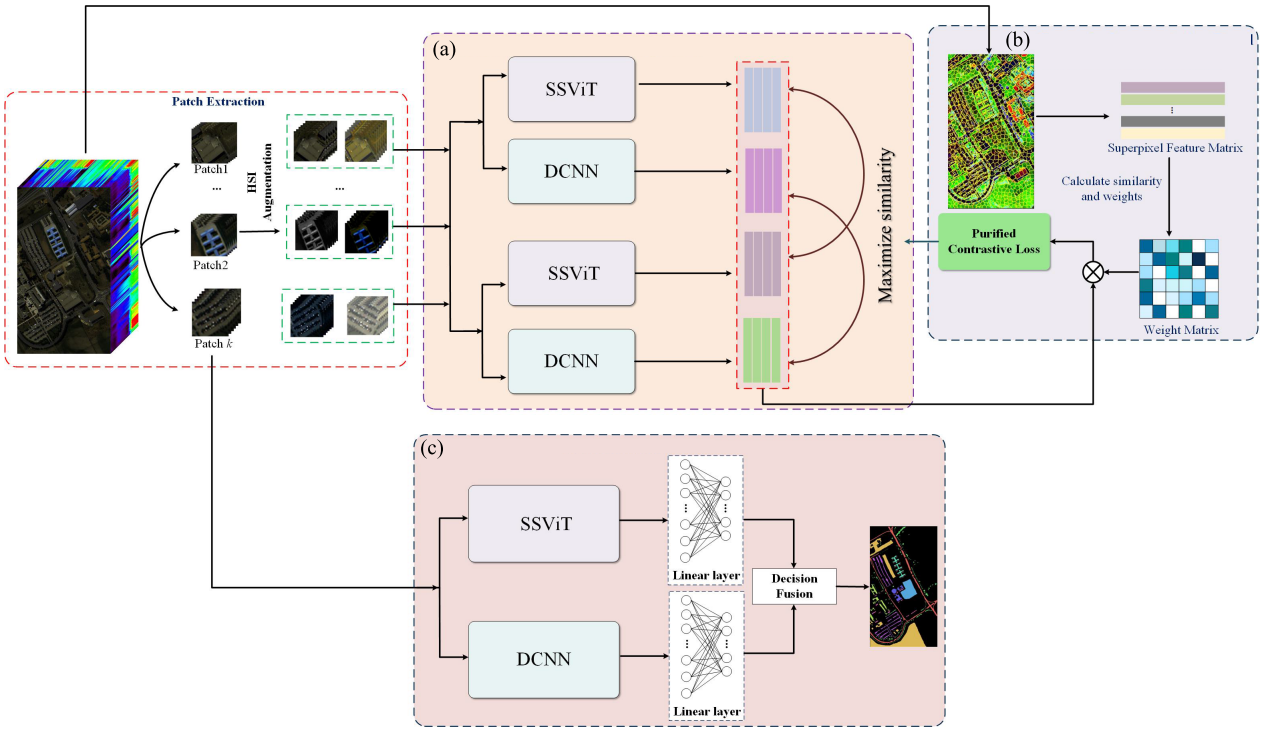

Fig. 2. Overview of the PCL framework. (a) Self-supervised pretraining model. (b) Superpixel segmentation-based purified contrastive loss. (c) Fine-tuning and classification model.

Part C focuses on network fine-tuning and the implementation of downstream classification tasks.

A. Spatial–Spectral Representation Encoder

An original HSI can be described as $X \in \mathbb{R}^{H\times W\times B}$, where H, W, and B are denoted as the height, width, and the number of spectral bands, respectively. Each pixel x in X has its corresponding land-cover class label $y \in \mathbb{R}^{1\times C}$, where C is the number of classes. First, the essential operation is to obtain two different views of the same image by data augmentation and form a positive sample pair, while the views from different images form a negative sample pair. In the previous work, we developed an integrated HSI data augmentation module and demonstrated the effectiveness of the module [30]. Therefore, the data augmentation module is adopted to obtain augmented views, incorporating operations, such as cropping, random color adjustment, and normalization. First, the segmented patches are first cropped to 0.95 times the original size and then resized to their original dimensions. This is followed by random adjustments in contrast and brightness within a narrow range. Finally, the pixel values within an image patch are normalized to range from 0 to 1. These augmented views are subsequently fed into a Siamese network structure-based contrastive learning framework, which consists of an SSViT subnetwork and a DCNN subnetwork.

1) SSViT Subnetwork: ViT is commonly used to capture global dependency information in the spatial dimension, but ignores the spectral dimension. HSIs are known for their high-resolution spectral information, which is beneficial to classification tasks. However, the local spectral information between different classes is very similar, which poses a challenge to effectively capture the interdependencies between long-range bands. Therefore, a spatial–spectral self-attention is designed to extract both global spatial and spectral information. The SSViT subnetwork structure, as shown in Fig. 3, consists of a subpatch embedding module, a spatial–spectral self-attention module, and a feed-forward module.

The subpatch embedding module consists primarily of a 3-D convolution unit and a 2-D convolution unit. Within the 3-D convolution unit, kernels of different sizes are used to extract correlation information within local spectral bands. The 2-D convolution unit is structured with two parallel convolutional subbranches, each incorporating three convolutional layers and two maximum pooling layers.

To effectively capture global dependencies in both spatial and spectral features, we introduce a spatial–spectral self-attention module. This module consists of a multihead spatial self-attention and a single-head spectral self-attention. First, the multihead spatial self-attention is used to model global spatial dependencies. Specifically, to derive representations of keys, values, and queries, we utilize convolutional projection with depthwise convolution and leaky rectified linear unit (Leaky ReLU) activations instead of traditional fully connected layers. We then perform a reshaping operation to transform the 3-D data into a 2-D format for subsequent self-attention computations. Formally, for an input feature map represented as $Z \in \mathbb{R}^{d\times h\times w}$, the respective query Q_i, key K_i, and value V_i are given by

$$\begin{aligned} Q_i &= \text{split}\big(\text{reshape}\big(DC_q(Z)\big)\big) \\ K_i &= \text{split}(\text{reshape}(DC_k(Z))) \\ V_i &= \text{split}(\text{reshape}(DC_v(Z))) \end{aligned} \tag{1}$$

where $i = 1, \ldots, s$, s is the number of self-attention heads; $DC_q(\cdot)$, $DC_k(\cdot)$, and $DC_v(\cdot)$ are the respective

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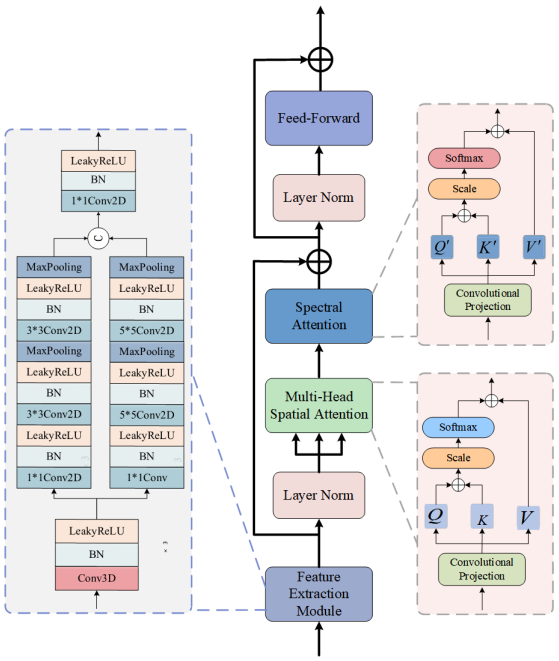

Fig. 3. SSViT subnetwork architecture.

depthwise convolutional operations with leaky ReLU nonlinearity; reshape(·) is the operation that converts a tensor with shape of $d \times h \times w$ into $s \times (h \times w) \times d_s$; and split(·) splits the tensor into s subtensors of the same size along the first dimension. Then, the self-attention can be expressed as follows:

$$\mathrm{SA}_i(Z) = \mathrm{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_s}}\right) V_i \tag{2}$$

where d_s equals d/s. The multihead spatial self-attention is an extension of SA, which is given by

$$\mathrm{MSA}_{\mathrm{spa}}(Z) = [\mathrm{SA}_1(Z), \mathrm{SA}_2(Z), \ldots, \mathrm{SA}_s(Z)] W_{\mathrm{spa}} \tag{3}$$

where $W_{\mathrm{spa}} \in \mathbb{R}^{d \times d}$ is a parameter matrix of linear projection.

Next, the output of multihead spatial self-attention is reshaped as a feature map $\tilde{Z} \in \mathbb{R}^{d \times h \times w}$ and fed into the spectral self-attention module for capturing global spectral correlation. Just as in the spatial self-attention mechanism, the query $\tilde{Q} \in \mathbb{R}^{d \times (h \times w)}$, key $\tilde{K} \in \mathbb{R}^{d \times (h \times w)}$, and value $\tilde{V} \in \mathbb{R}^{d \times (h \times w)}$ are obtained by depthwise convolutions and reshaping operations. Subsequently, the spectral self-attention is defined as follows:

$$\mathrm{SA}_{\mathrm{spe}}(\tilde{Z}) = \mathrm{softmax}\left(\frac{\tilde{Q}\tilde{K}^T}{\sqrt{d_s}}\right) \tilde{V}. \tag{4}$$

The major characteristic of the spatial–spectral self-attention is that depthwise convolution can effectively keep the independence between the spatial and spectral information while modeling the correlation of spatial pixels and spectral bands. Furthermore, the number of parameters of depthwise convolution is significantly lower compared with commonly used fully connected layers.

2) DCNN Subnetwork: Despite the fact that the ViT can focus on global correlations, it is susceptible to ignore structural information. To effectively model local contextual information in spatial dimensions, an encoder based on ResNet-18 [44] is used to extract spatial structural features. The ResNet family of networks are effective in avoiding model degradation problems because of its unique residual structure and have stronger robustness. The number of input channels and the size of the convolution kernel in the first convolution layer are modified to make the input of ResNet suitable for the HSI, and the last FC layer is removed. In addition, an additional projection layer, which contains two linear layers and a ReLU activation function, is added to the encoder.

B. Purified Contrastive Loss

After completing the feature extraction in Section III-A, the subsequent procedure involves computing the contrastive loss. The most commonly used loss function for contrastive learning is the InfoNCE [45], but it fails to distinguish between true negative samples and false negative samples. To alleviate the negative impact of the false negative samples on the contrastive learning, a "clearer" contrastive loss is obtained by assigning the weights to negative sample pairs based on their similarity, which is termed "purified contrastive loss." This loss effectively reduces the influence of false negative sample pairs by giving the lower weights to the sample pairs that exhibit greater similarity. Fig. 4 illustrates the workflow of the purified InfoNCE contrastive loss based on superpixel segmentation.

First, we use a simple linear iterative clustering (SLIC) [46] to obtain the superpixel map S from HSI X. The number of superpixels is expressed as $K = \lceil (W \times H)/a \rceil$, where a denotes the average segmentation size of each superpixel and the operator $\lceil \cdot \rceil$ represents round toward positive infinity. According to the above operations, the pixel-level regular spatial structure information is mapped to the irregular superpixel region, and an association matrix $U \in \mathbb{R}^{(W \times H) \times K}$ is constructed. This matrix facilitates the accurate indexing of pixels to their corresponding superpixels and enables the transformation of features between pixels and superpixels [47]. The association matrix is given by

$$U_{i,k} = \begin{cases} 1, & if\ x_i \in S_k \\ 0, & if\ x_i \notin S_k \end{cases} \tag{5}$$

where x_i represents the ith pixel in HSI X. The superpixel spectral feature matrix $F \in \mathbb{R}^{K \times B}$ can be defined by averaging the signatures of pixels belonging to the same superpixel region. The spectral feature of the kth superpixel is generated as follows:

$$F_k = \frac{1}{T} \sum_{i=1}^{T} x_k^i \tag{6}$$

where x_k^i denotes the ith pixel signature in the kth superpixel S_k and T is the number of pixels located in the kth superpixel region.

Through data augmentation operations, $2N$ augmented samples $\{\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n, \tilde{x}_{n+1}, \ldots, \tilde{x}_{2N}\}$ will be obtained from a mini-batch of size N. Each sample $\tilde{x}_n$ together with the other augmented samples forms $2N$-1 sample pairs, consisting of one positive sample pair and $2N$-2 negative sample pairs. To alleviate the effect of false negative samples, it is necessary

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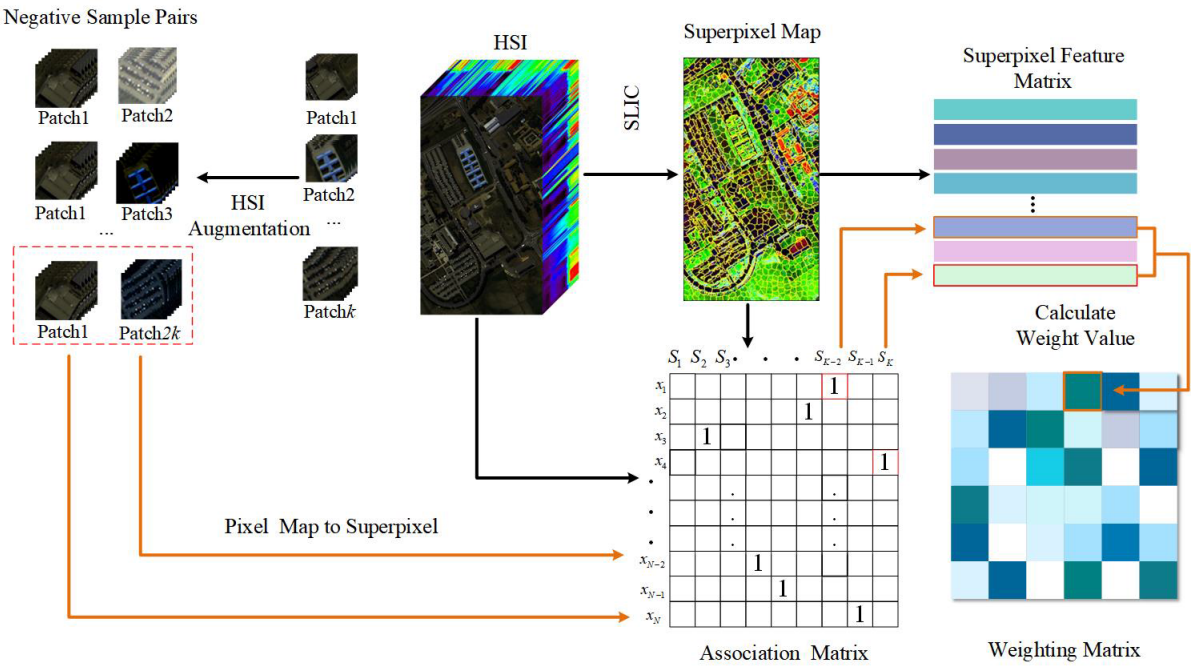

Fig. 4. Workflow of negative sample pair weighting based on superpixel segmentation.

to first quantify the similarity between the two samples of a negative sample pair. Given the augmented samples $\tilde{x}_j$ and $\tilde{x}_n$, their features can be derived from the corresponding superpixel by

$$F_j = \tilde{U}_j \times F, \quad j = 1, \ldots 2N$$
$$F_n = \tilde{U}_n \times F, \quad n = 1, \ldots 2N \tag{7}$$

where $\tilde{U}_j$ and $\tilde{U}_n$ are the jth and the nth row of U, respectively, and $\tilde{U}$ is extended from U due to sample data augmentation, such that $\tilde{U}_{2i-1} = \tilde{U}_{2i} = U_i$, $i = 1, \ldots, N$.

Subsequently, the cosine similarity function is introduced to calculate the spectral similarity of the two samples, as follows:

$$\mathrm{Sim}(F_j, F_n) = \frac{F_j^T \cdot F_n}{\|F_j\| \times \|F_n\|}. \tag{8}$$

The above procedures elegantly compute the similarity of samples within a negative sample pair. Then, the weight of each negative sample pair is given by

$$w_{j,n} = \alpha^{\frac{\mathrm{sim}(F_j, F_n)}{\beta}}, \quad 0 < \alpha < 1, 0 < \beta < 1 \tag{9}$$

where α and β are used to regulate the range of dilation and diminution. By default, α and β are set to 0.75 and 0.25, respectively. The negative sample pairs with high similarity obtain a smaller weight value, while, conversely, the negative sample pairs with low similarity obtain a larger weight value adaptively. The generated weights will be multiplied with the values in the denominator of the InfoNCE loss function. The purified contrastive loss function for the given augmented sample $\tilde{x}_j$ can be expressed as follows:

$$\mathrm{Loss}_j = -\log \frac{\exp\left(\mathrm{sim}\left(z_j, z_{\tilde{j}}\right)/\tau\right)}{\sum_{n=1}^{2N} \mathbb{I}\left[n \neq j\right] w_{j,n} \times \exp\left(\mathrm{sim}\left(z_j, z_n\right)/\tau\right)} \tag{10}$$

where z_j and $z_{\tilde{j}}$ are the learned representations from the same positive pair, τ is a temperature parameter and is set to 0.5 by default, $w_{j,n}$ is the weight value of each negative sample pair, and $\mathbb{I}_{[n \neq j]} \in \{0, 1\}$ is an indicator function when $n \neq j$ the evaluation is 1. In summary, the proposed purified InfoNCE contrastive loss can adaptively adjust the weight value of each negative sample pairs, which can effectively alleviate the effect of false negative sample pairs and promote the model for better contrastive training.

To better optimize the hybrid SSViT-DCNN encoder, a joint loss strategy is proposed. The joint loss can be formulated as follows:

$$\mathrm{Loss}_{\mathrm{total}} = \mathrm{Loss}\left(s_j, s_{\tilde{j}}\right) + \mathrm{Loss}\left(r_j, r_{\tilde{j}}\right) \tag{11}$$

where s_j and $s_{\tilde{j}}$, and r_j and $r_{\tilde{j}}$ are the learned representations through SSViT and DCNN networks for a positive pair of samples $\tilde{x}_j$ and $\tilde{x}_{\tilde{j}}$.

The benefit of the joint loss is that the model does not optimize the DCNN and the SSViT separately. By combining the losses from these two branches, the network attempts to minimize their sum during backpropagation. This combination between branches allows for collaborative adjustments in feature representations across branches, effectively minimizing the total loss and consequently leading to improved performance.

C. Fine-Tuning by Limited Labeled Samples

After completing the pretraining process, the DCNN-SSViT encoder is transferred to the downstream classification task, as shown in Fig. 2. Following the SSViT and DCNN subnetworks, two fully connected layers are added to serve as classifiers. With limited labeled samples, the pretrained model parameters are fine-tuned, and the linear layers of the classifiers are trained in a supervised learning manner.

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TABLE I
CONFIGURATION OF THE TRAINING AND TESTING SAMPLE NUMBERS OF THE PAVIA UNIVERSITY DATASET, INDIA PINES DATASET, AND HOUSTON 2013 DATASET

Dataset	Pavia University dataset			Indian Pines dataset			Houston 2013 dataset		
No.	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Asphalt	5	6626	Alfalfa	5	41	Healthy grass	5	1246
2	Meadows	5	18644	Corn-notill	5	1423	Stressed grass	5	1249
3	Gravel	5	2094	Corn-mintill	5	825	Synthetic grass	5	692
4	Trees	5	3059	Corn	5	232	Trees	5	1239
5	Metal Sheets	5	1340	Grass- pasture	5	478	Soil	5	1237
6	Bare soil	5	5024	Grass-trees	5	7	Water	5	320
7	Bitumen	5	1325	Grass-pasture- mowed	5	23	Residential	5	1263
8	Bricks	5	3677	Hay-windrowed	5	473	Commercial	5	1239
9	Shadows	5	942	Oats	5	15	Road	5	1247
10				Soybean-notill	5	967	Highway	5	1222
11				Soybean-mintill	5	2450	Railway	5	1230
12				Soybean-clean	5	588	Parking Lot 1	5	1228
13				wheat	5	200	Parking Lot 2	5	464
14				woods	5	1260	Tennis Court	5	423
15				Buildings	5	381	Running Track	5	655
16				Stone	5	88			
Total		45	42731		80	10169		75	14954

During the final testing stage, the prediction results of the SSViT and DCNN subnetworks are jointly used to determine the land-cover class through decision-level fusion. The final prediction result is expressed as follows:

$$Y_{\text{out}} = \eta Y_{\text{SSViT}} + (1-\eta) Y_{\text{DCNN}} \tag{12}$$

where η is a hyperparameter, and Y_{SSViT} and Y_{DCNN} represent the prediction results from the SSViT and the DCNN subnetworks, respectively.

III. EXPERIMENTAL RESULT AND ANALYSIS

A. HSI Dataset

To demonstrate the effectiveness of our proposed method, we conducted a series of experiments on three public datasets, which included Pavia University, Indian Pines, and Houston 2013.

The University of Pavia scene dataset (UP) was taken by ROSIS of Germany on the city of Pavia, Italy, which is part of the hyperspectral data of the Pavia University. The size of this data is 610 $\times$ 340 with 103 spectral bands, and the spectral coverage ranges from 0.43 to 0.86 μm. Besides, it contains a total of 42 776 labeled pixels with nine classes of interest.

The Indian Pines scene dataset (IP) is an image of an Indian pine area in Indiana, USA, in 1992, by the Airborne Visual Infrared Imaging Spectrometer (AVIRIS). The image has 200 spectral bands of size 145 $\times$ 145 with 16 ground-truth classes, and the wavelength range is from 0.4 to 2.5 μm.

The Houston 2013 dataset (HU) was collected by the ITRES CASI-1500 sensor and previously applied in the 2013 GRSS data fusion competition. It contains information about the University of Houston campus and its surrounding community. The dataset consists of 349 $\times$ 1905 pixels and includes 15 classes of ground objects, while having 144 spectral bands ranging from 0.38 to 1.05 μm. More details about the land-cover classes, the number of training samples, and the number of test samples for different datasets are shown in Table I.

B. Experimental Set

1) Evaluation Indexes: To measure the classification performance of different methods, the overall accuracy (OA), average accuracy (AA), and Kappa are introduced. The OA denotes the number of correctly classified samples among the total test samples, while AA is the average of the sum for each class of classification accuracy. The Kappa is an indication of the consistency between the ground truth and the classification map.

2) Parameter Settings: For each dataset, we randomly selected 50% of the data as the unlabeled training dataset for the contrastive learning pretext task. During supervised fine-tuning, we randomly selected five labeled samples per class as labeled training samples and the rest as test samples. The size of each hyperspectral patch was set to 15 $\times$ 15, and the Adam optimizer was applied to update the model parameters during both pretraining and fine-tuning. In the pretraining step, the learning rate (lr) was set to $1e-4$, while the epoch and batch size were set to 400 and 256, respectively. In the fine-tuning step, we set the epoch to 300 and the lr of the classifier to $1e-3$. In addition, we used a smaller lr to fine-tune the pretrained model on different datasets: lr $\times$ 0.02 for UP and lr $\times$ 0.01 for both IP and HU.

To demonstrate the effectiveness of our method, we conducted comparative experiments with state-of-the-art classification methods. These include supervised methods, such as SVM-RBF [48], 2D-CNN [49], HiT [50], ViT [51], SSFTT [52], AMGCFN [53], and DM-MRN [54], as well as self-supervised methods, such as the L2CSL [30], the C-SS-MTr [55], and the semi-supervised method Res-CP [56]. To ensure fairness, five labeled samples per class were randomly selected for training in each trial. Specifically, for the supervised learning and the semi-supervised methods, the model was trained and tested ten times. For the self-supervised methods, the model was pretrained once, and the fine-tuning and testing of the pretrained model were conducted ten times.

The experimental computer setup includes an Intel[1] Core[2] i7-10700F processor, an NVIDIA GeForce RTX 2060 SUPER graphics card, and Python version 3.8.5 and PyTorch version

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TABLE II
CLASSIFICATION PERFORMANCE IN TERMS OF OA, AA, AND KAPPA USING DIFFERENT HEAD NUMBERS

	2	4	8
OA(%)	73.62	74.31	73.24
AA(%)	85.24	86.16	85.23
$k\times100$	70.36	70.93	69.81

TABLE III
CLASSIFICATION PERFORMANCE IN TERMS OF OA, AA, AND KAPPA USING DIFFERENT DEPTHS

	2	4	6
OA(%)	74.87	72.95	76.71
AA(%)	86.88	85.23	87.03
$k\times100$	71.87	69.79	73.85

TABLE IV
CLASSIFICATION PERFORMANCE OF PCL WITH DIFFERENT FUSION COEFFICIENTS η

	0.2	0.4	0.6	0.8
OA(%)	69.71	72.58	74.82	76.71
AA(%)	82.72	84.95	85.53	87.03
$k\times100$	66.06	69.33	71.72	73.85

1.7.1 as the programming language and deep learning platform, respectively.

3) Parameter Analysis: Although most of the parameters used in our experiments were set empirically by default, we conducted experiments to determine the final values for some important parameters. For the fine-tuning procedure, labeled samples from each class randomly were selected as training samples. We first evaluated the classification performance of our model on the IP dataset by testing different numbers of heads and depths. Specifically, we tested head numbers of 2, 4, and 8, and depths of 2, 4, and 6. Table II illustrates that the highest OA is achieved when the number of heads was set to 4. Similarly, Table III shows that the best OA is achieved when the depth of each head was set to 6. In addition, we also investigated the optimal value for the hyperparameter η, which comes from a candidate set of 0.2, 0.4, 0.6, and 0.8. Since different subbranches contribute differently to the final classification results, it is essential to consider the optimal value of η. Table IV indicates that the model achieves the highest classification accuracy when the value of η is set to 0.8. Finally, to examine the impact of patch size on classification results on different datasets, we tested the performance under different patch sizes of 9×9, 11×11, 13×13, 15×15, and 17×17 and recorded the OA. As shown in Fig. 5, the 15×15 patch size consistently yielded higher OA for all datasets. Therefore, we selected the 15×15 patch size as the optimal patch size.

C. Analysis of Experimental Result

1) Quantitative Analysis: Tables V–VII show the classification performance of the different methods on the UP, IP, and HU datasets, where the best result for each row is represented in bold font in the table. All results are listed in the form of mean value ± standard deviations. From the results in Tables V–VII, it can be clearly observed that the self-supervised learning classification methods, such as L2CSL, C-SS-MTr, and our method, have better classification performance compared with other supervised learning methods. This is because supervised learning methods have remarkable classification performance only with sufficient labeled samples, but when labeled samples are extremely scarce, these methods are very prone to overfitting.

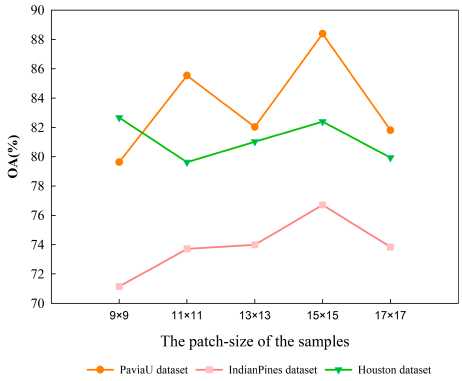

Fig. 5. OA under different patch sizes.

Specifically, on the UP dataset, our method achieves competitive classification results; among the self-supervised learning methods, the OA of our method (88.39%) is higher than L2CSL by 2.86%. Compared with supervised learning methods, our method is nearly 4.75% higher than the best method AMGCFN among them. Notably, attributed to the ability of the Transformer-based model to capture global information, the AMGCFN (83.64%) has better OA compared with other supervised learning methods. On the IP dataset, it can be seen from Table VI that the proposed method achieves the highest values of the evaluation metrics. The values of OA, AA, and Kappa are 76.71%, 87.03%, and 73.85%, respectively, which are improved by 4.42%, 3.82%, and 7.64% compared with the best self-supervised method C-SS-MTr. The value of OA is increased by 1.23% compared with the best supervised model AMGCFN among supervised learning methods. It demonstrates the effectiveness of supervised learning in hyperspectral classification task with limited labeled samples. Meanwhile, we also note that the L2CSL and C-SS-MTr fail to achieve the satisfactory classification results. The reason is that the spatial structure of the IP dataset is extremely similar, and the negative sample conflict phenomenon is highly likely to occur after the sampling is completed, which further leading to a failure of the contrast training. The proposed weighted loss can effectively alleviate this problem to promote the model for better contrastive training, thus achieving the best results. In Table VII, the OA of the proposed method (82.40%) on the HU dataset is higher 2.95% than that of C-SS-MTr (79.45%). Besides, compared with the best supervised learning method, the values of OA, AA, and Kappa are increased by 5.54%, 4.76%, and 5.96%, respectively. In summary, the classification results on the three datasets demonstrate the effectiveness of our method,

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TABLE V
TESTING ACCURACIES OF DIFFERENT MODELS IN UP DATASET

Class No.	SVM-RBF	2D-CNN	HiT	ViT	SSFTT	AMGCFN	DM-MRN	Res-CP	L2CSL	C-SS-MTr	PCL
1	69.12 (±6.27)	69.37 (±1.57)	60.49 (±2.22)	59.70 (±6.54)	65.00 (±5.38)	81.44 (±3.03)	76.30 (±6.96)	**94.85 (±1.17)**	80.13 (±4.93)	85.42 (±3.21)	90.61 (±2.86)
2	78.26 (±2.86)	82.73 (±4.38)	65.85 (±4.53)	84.76 (±7.35)	88.25 (±3.01)	80.75 (±5.59)	80.85 (±8.43)	**99.42 (±0.00)**	92.29 (±2.93)	73.35 (±2.76)	91.28 (±2.92)
3	44.22 (±6.73)	49.20 (±5.95)	40.27 (±11.04)	61.50 (±7.27)	41.71 (±4.65)	83.61 (±6.25)	77.70 (±2.34)	71.12 (±7.27)	73.77 (±10.00)	**86.38 (±2.08)**	81.63 (±5.89)
4	65.36 (±6.32)	85.37 (±7.90)	89.15 (±1.04)	68.42 (±6.18)	84.12 (±7.23)	74.01 (±1.83)	**93.40 (±6.50)**	49.29 (±9.56)	88.31 (±3.01)	71.92 (±11.54)	77.89 (±12.18)
5	94.94 (±3.77)	95.40 (±2.23)	98.97 (±0.11)	93.88 (±2.11)	94.17 (±2.46)	99.28 (±0.05)	**99.44 (±0.02)**	85.24 (±11.73)	93.21 (±2.62)	98.96 (±1.20)	97.62 (±2.12)
6	41.66 (±12.44)	57.03 (±7.23)	58.82 (±11.32)	70.62 (±10.97)	50.22 (±10.97)	89.59 (±1.86)	**93.66 ±6.53**	60.34 (±13.52)	80.25 (±4.43)	93.06 (±2.51)	88.46 (±6.07)
7	50.38 (±8.41)	56.10 (±3.33)	63.61 (±6.53)	51.82 (±4.14)	43.59 (±9.24)	97.68 (±1.11)	**99.50 (±0.07)**	55.88 (±9.24)	85.01 (±3.99)	96.89 (±0.31)	91.38 (±5.59)
8	63.76 (±4.65)	58.03 (±7.28)	67.20 (±5.22)	66.28 (±4.29)	63.76 (±7.68)	90.64 (±4.00)	68.68 (±1.89)	85.01 (±4.01)	78.60 (±5.06)	**92.70 (±3.34)**	86.27 (±5.71)
9	**99.84 (±0.08)**	77.70 (±5.38)	96.54 (±0.11)	51.40 (±9.48)	74.35 (±12.55)	86.57 (±7.84)	97.62 (±0.85)	99.82 (±0.01)	79.03 (±6.90)	65.35 (±11.32)	87.35 (±9.43)
OA(%)	68.32 (±2.62)	72.71 (±3.28)	66.36 (±3.29)	72.71 (±5.43)	73.76 (±1.60)	83.64 (±3.29)	82.88 (±4.72)	78.87 (±1.17)	85.53 (±2.55)	81.10 (±4.49)	**88.39 (±2.35)**
AA(%)	72.68 (±1.76)	74.44 (±3.00)	71.21 (±1.98)	75.19 (±2.34)	70.52 (±1.90)	87.06 (±1.94)	87.40 (±3.53)	73.87 (±2.23)	87.41 (±1.53)	84.89 (±3.17)	**91.65 (±1.13)**
$\kappa\times100$	59.10 (±3.44)	64.90 (±4.00)	58.32 (±3.33)	65.55 (±6.01)	65.33 (±1.84)	79.05 (±3.95)	78.23 (±5.73)	77.91 (±3.22)	81.28 (±3.07)	76.24 (±4.86)	**85.10 (±2.85)**

TABLE VI
TESTING ACCURACIES OF DIFFERENT MODELS IN IP DATASET

Class No.	SVM-RBF	2D-CNN	HiT	ViT	SSFTT	AMGCFN	DM-MRN	Res-CP	L2CSL	C-SS-MTr	PCL
1	32.77 (±7.62)	75.92 (±11.05)	98.89 (±0.03)	42.82 (±12.76)	49.66 (±19.35)	98.29 (±4.37)	98.54 (±0.24)	52.56 (±10.22)	38.75 (±13.47)	**98.78 (±0.14)**	72.60 (±13.75)
2	38.57 (±8.58)	45.56 (±9.88)	32.37 (±11.92)	44.86 (±15.69)	50.22 (±7.96)	59.00 (±13.28)	43.34 (±8.45)	64.75 (±11.24)	61.61 (±4.62)	52.82 (±6.03)	**67.13 (±3.89)**
3	29.67 (±5.97)	54.24 (±5.91)	29.02 (±12.54)	48.96 (±8.04)	53.96 (±7.69)	73.67 (±5.49)	52.52 (±11.81)	**75.67 (±4.72)**	53.38 (±5.00)	59.96 (±12.89)	74.48 (±10.73)
4	34.53 (±6.62)	55.82 (±13.78)	64.58 (±9.43)	59.70 (±7.73)	54.52 (±13.51)	90.82 (±9.93)	90.34 (±5.19)	59.24 (±11.21)	69.30 (±12.79)	**91.47 (±1.74)**	73.27 (±9.45)
5	49.93 (±8.49)	47.54 (±11.15)	43.47 (±8.92)	62.42 (±18.49)	78.46 (±6.00)	73.75 (±7.03)	73.58 (±7.50)	79.64 (±5.90)	74.08 (±4.22)	76.97 (±4.12)	**84.52 (±6.94)**
6	75.47 (±5.95)	74.11 (±4.96)	80.03 (±1.32)	87.80 (±4.33)	87.22 (±3.64)	85.54 (±5.83)	94.51 (±2.49)	94.02 (±0.89)	87.65 (±5.50)	89.57 (±3.54)	**95.11 (±2.79)**
7	37.30 (±7.44)	44.20 (±15.06)	98.89 (±0.03)	29.24 (±3.00)	38.76 (±4.39)	92.44 (±0.64)	**99.57 (±0.01)**	29.96 (±13.24)	46.66 (±7.94)	97.83 (±1.11)	35.26 (±8.45)
8	73.13 (±10.43)	96.38 (±2.07)	82.01 (±7.21)	95.34 (±2.04)	89.18 (±12.79)	**99.79 (±0.28)**	99.24 (0.05)	80.12 (±4.01)	90.89 (±4.39)	99.34 (±0.03)	97.27 (±2.06)
9	19.13 (±6.06)	21.97 (±2.16)	**100.00 (±0.00)**	20.42 (±4.49)	19.26 (±7.33)	96.03 (±2.56)	91.03 (±2.56)	36.59 (±5.22)	36.89 (±12.67)	42.51 (±5.76)	35.86 (±17.18)
10	35.60 (±5.74)	44.34 (±7.03)	57.28 (±3.65)	59.60 (±5.86)	59.22 (±4.73)	**74.20 (±7.32)**	61.44 (±12.48)	56.72 (±6.31)	65.33 (±11.82)	72.99 (±5.43)	67.73 (±6.25)
11	45.53 (±5.20)	55.42 (±6.28)	45.37 (±4.01)	66.70 (±4.36)	63.24 (±4.23)	67.69 (±6.71)	57.75 (±6.16)	**98.51 (±0.04)**	68.72 (±9.92)	64.13 (±4.67)	72.87 (±5.56)
12	23.67 (±5.14)	44.02 (±8.27)	49.95 (±5.42)	36.42 (±6.95)	53.02 (±5.14)	**70.18 (±12.78)**	51.79 (±5.69)	52.67 (±5.53)	60.11 (±5.08)	59.39 (±7.62)	58.94 (±9.36)
13	75.77 (±4.23)	81.61 (±8.08)	96.31 (±1.13)	70.58 (±9.22)	73.96 (±4.12)	96.15 (±5.09)	**96.85 (±1.09)**	81.97 (±6.22)	81.16 (±6.25)	95.72 (±0.41)	88.52 (±4.83)
14	72.07 (±7.90)	79.86 (±7.21)	82.36 (±3.12)	87.62 (±5.03)	90.24 (±3.38)	86.03 (±9.45)	93.69 (±2.46)	86.45 (±3.11)	84.49 (±2.65)	88.59 (±3.41)	**94.41 (±2.30)**
15	29.30 (±3.90)	71.43 (±8.39)	34.41 (±11.67)	67.96 (±7.98)	76.34 (±4.48)	90.65 (±6.85)	77.61 (±5.19)	52.56 (±10.29)	70.03 (±7.51)	**85.38 (±2.67)**	83.67 (±5.01)
16	84.73 (±7.84)	66.76 (±12.74)	99.28 (±0.01)	54.76 (±12.44)	63.62 (±17.58)	95.41 (±1.45)	96.82 (±1.42)	69.29 (±11.32)	46.93 (±14.47)	**98.41 (±0.01)**	67.54 (±5.66)
OA(%)	46.65 (±2.19)	58.74 (±1.95)	54.09 (±1.74)	64.22 (±3.20)	66.13 (±2.81)	75.48 (±2.14)	67.98 (±3.25)	72.57 (±2.56)	68.96 (±3.98)	72.47 (±4.36)	**76.71 (±3.04)**
AA(%)	58.86 (±1.97)	73.68 (±1.02)	68.39 (±1.31)	76.71 (±3.01)	78.88 (±2.47)	85.07 (±1.39)	80.47 (±2.03)	69.54 (±2.56)	81.61 (±1.73)	83.21 (±2.84)	**87.03 (±2.19)**
$\kappa\times100$	40.41 (±2.07)	54.01 (±1.77)	49.29 (±1.28)	60.00 (±3.57)	62.14 (±3.17)	72.43 (±2.67)	64.20 (±3.49)	66.91 (±2.31)	65.41 (±4.23)	36.21 (±5.15)	**73.85 (±3.39)**

which has better classification performance compared with other methods under the condition of extremely limited labeled samples (only five labeled samples).

2) Classification Maps: For a more intuitive visual evaluation, we visualized the classification results of each method that are consistent with the results in Tables V–VII. Figs. 6–8 show the classification maps of different methods on the University of Pavia, Indian Pines, and Houston 2013 datasets, respectively. Visually comparing with other approaches, our method consistently produces the most accurate and smooth classification maps across three distinct datasets. Using the University of the Indian Pines dataset as an example, our method achieves the highest prediction accuracy despite the fact that the *Corn-notill* class contains numerous irregularly shaped parts and is unevenly distributed. Similarly on the Houston 2013 dataset, the *Syntheticgrass* class has a more competitive classification accuracy than other methods. It is also noteworthy that self-supervised learning-based methods, such as L2CSL and C-SS-MTr, also have less noisy of classification maps on the three datasets due to their powerful ability to extract generalized representations. Other methods, such as 2DCNN, ViT, and SSFTT, demonstrate close

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TABLE VII
TESTING ACCURACIES OF DIFFERENT MODELS IN HU DATASET

Class No.	SVM-RBF	2D-CNN	HiT	ViT	SSFTT	AMGCFN	DM-MRN	Res-CP	L2CSL	C-SS-MTr	PCL
1	**89.23 (±2.83)**	77.88 (±2.81)	87.86 (±1.91)	73.68 (±3.63)	81.6 (±4.46)	86.15 (±6.33)	77.39 (±5.21)	80.9 (±5.21)	78.38 (±5.03)	81.04 (±4.78)	84.16 (±2.74)
2	84.93 (±3.87)	68.28 (±5.06)	62.03 (±3.30)	66.68 (±1.88)	75.0 (±4.93)	69.22 (±4.31)	86.10 (±6.69)	73.16 (±7.14)	72.34 (±10.87)	73.10 (±8.45)	**88.38 (±2.14)**
3	81.80 (±11.89)	93.91 (±15.72)	86.04 (±7.94)	91.40 (±7.56)	82.44 (±16.46)	97.90 (±2.17)	97.79 (±1.56)	96.66 (±2.21)	93.38 (±4.64)	96.04 (±1.15)	**98.08 (±1.79)**
4	**94.10 (±2.21)**	74.47 (±13.92)	85.49 (±2.21)	60.38 (±4.21)	78.98 (±5.46)	75.88 (±7.29)	89.21 (±4.28)	84 (±3.79)	82.59 (±8.59)	82.23 (±6.13)	89.77 (±2.95)
5	88.03 (±6.78)	93.72 (±3.35)	82.82 (±7.42)	93.30 (±2.36)	88.68 (±3.36)	96.55 (±5.23)	96.10 (±7.27)	98.18 (±0.02)	87.66 (±4.90)	**98.54 (±0.03)**	94.75 (±1.88)
6	82.83 (±7.81)	84.43 (±6.04)	78.76 (±5.23)	73.58 (±7.16)	68.0 (±10.59)	87.90 (±5.97)	**89.50 (±7.93)**	88.72 (±3.21)	78.59 (±4.93)	88.65 (±3.67)	84.79 (±6.80)
7	61.17 (±6.93)	62.26 (±12.97)	73.40 (±2.66)	63.20 (±6.52)	70.32 (±4.80)	76.53 (±1.03)	63.03 (±9.04)	**79.19 (±2.33)**	83.55 (±3.01)	78.14 (±5.22)	75.84 (±7.89)
8	48.70 (±6.42)	43.46 (±10.12)	46.60 (±10.64)	60.10 (±4.39)	47.2 (±13.42)	41.51 (±11.88)	37.28 (±10.78)	54.77 (±7.93)	55.21 (±8.71)	56.68 (±9.41)	**65.81 (±8.75)**
9	58.53 (±4.35)	60.38 (±8.53)	69.26 (±7.22)	66.94 (±3.30)	69.24 (±5.83)	56.69 (±4.88)	61.51 (±6.97)	77.82 (±2.47)	71.59 (±8.73)	**78.24 (±4.67)**	73.08 (±7.80)
10	42.97 (±9.29)	54.28 (±10.45)	62.23 (±9.54)	51.42 (±3.47)	60.6 (±6.70)	78.24 (±3.36)	70.62 (±9.29)	**81.64 (±1.35)**	71.54 (±9.23)	80.48 (±4.51)	73.93 (±3.25)
11	44.20 (±16.86)	66.98 (±7.16)	56.92 (±8.62)	64.22 (±1.96)	70.08 (±5.47)	70.77 (±9.73)	77.55 (±3.61)	74.86 (±6.78)	77.79 (±4.71)	74.65 (±6.74)	**81.06 (±5.99)**
12	40.13 (±9.71)	55.64 (±8.52)	63.20 (±7.94)	66.94 (±2.49)	58.54 (±8.83)	**80.45 (±7.95)**	67.66 (±11.32)	67.07 (±6.01)	69.78 (±5.57)	62.7 (±6.53)	72.18 (±6.81)
13	12.50 (±3.26)	75.98 (±9.26)	**93.14 (±1.23)**	57.52 (±10.24)	74.34 (±11.04)	89.24 (±6.28)	88.97 (±3.89)	84.74 (±3.32)	86.42 (±5.03)	81.78 (±5.15)	82.51 (±11.60)
14	80.70 (±8.24)	88.96 (±6.57)	85.84 (±5.34)	98.68 (±0.58)	88.58 (±3.92)	99.76 (±0.02)	99.59 (±0.06)	**99.88 (±0.00)**	95.78 (±2.57)	96.54 (±2.42)	98.07 (±1.93)
15	98.07 (±0.25)	85.57 (±8.06)	83.58 (±4.92)	91.90 (±4.86)	87.58 (±9.97)	95.35 (±1.49)	**98.37 (±0.36)**	95.53 (±1.23)	86.55 (±8.31)	96.82 (±0.58)	94.25 (±9.65)
OA(%)	67.98 (±2.30)	69.58 (±2.71)	71.86 (±1.99)	70.09 (±1.08)	72.61 (±3.22)	76.86 (±4.27)	76.56 (±2.56)	80 (±1.21)	78.02 (±2.78)	79.45 (±2.35)	**82.40 (±1.88)**
AA(%)	69.48 (±2.20)	72.69 (±2.34)	74.48 (±0.24)	73.08 (±0.98)	75.59 (±3.64)	80.14 (±3.61)	80.05 (±2.24)	82.48 (±2.02)	80.76 (±2.99)	81.94 (±2.01)	**84.90 (±1.87)**
$\kappa\times100$	65.43 (±2.49)	67.11 (±2.91)	69.7 (±1.71)	67.70 (±1.15)	70.46 (±3.51)	75.03 (±4.58)	74.71 (±2.79)	78.4 (±2.33)	77.39 (±3.50)	77.82 (±2.53)	**80.99 (±2.02)**

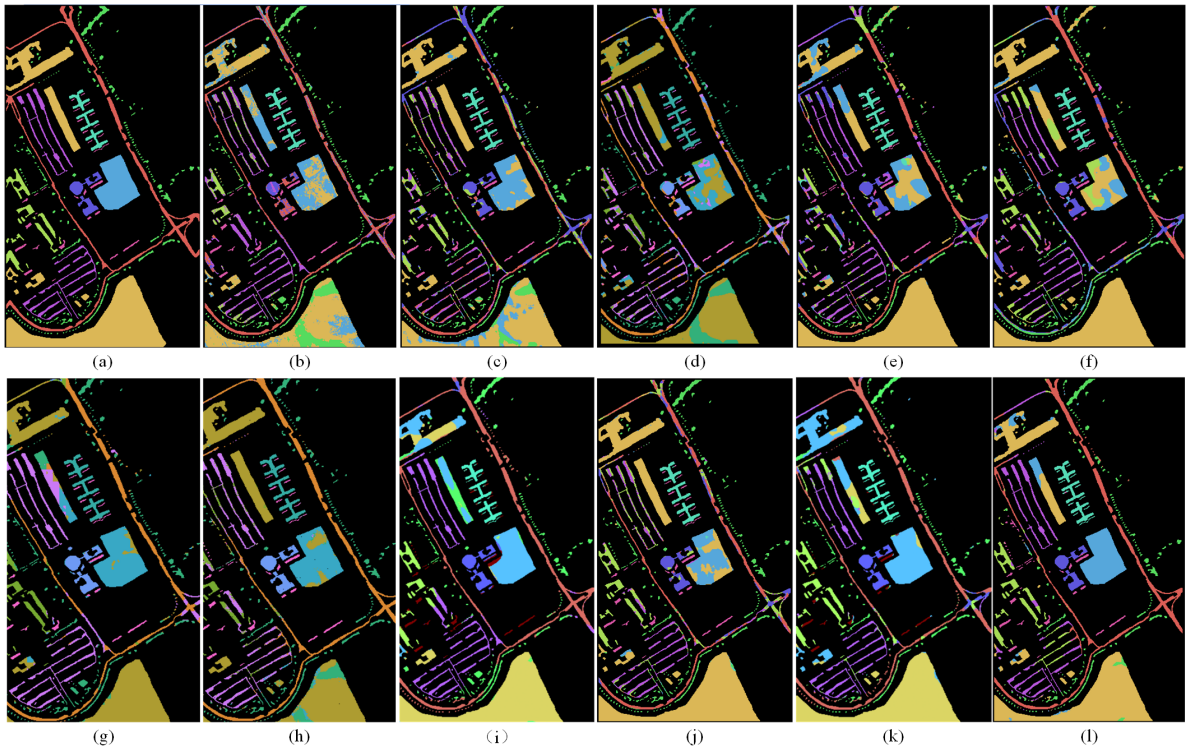

Fig. 6. Classification map of the University of Pavia dataset. (a) Ground truth. (b) SVM-RBF (68.32%). (c) 2DCNN (73%). (d) HiT (66.36%). (e) ViT (72.71%). (f) SSFTT (73.76%). (g) AMGCFN (83.64%). (h) DM-MRN (82.88%). (i) Res-CP (78.87%). (j) L2CSL (85.53%). (k) C-SS-MTr (81.10%). (l) PCL (88.39%).

classification results due to the difficulty in fully exploiting their performance under the condition of limited labeled samples. However, the traditional method SVM-RBF exhibits significant noise in the visualization results on the three datasets. Especially on the Indian Pines dataset, the noise of the classification map is more obvious. This also indirectly indicates that the traditional method is difficult to identify the land-cover classes accurately.

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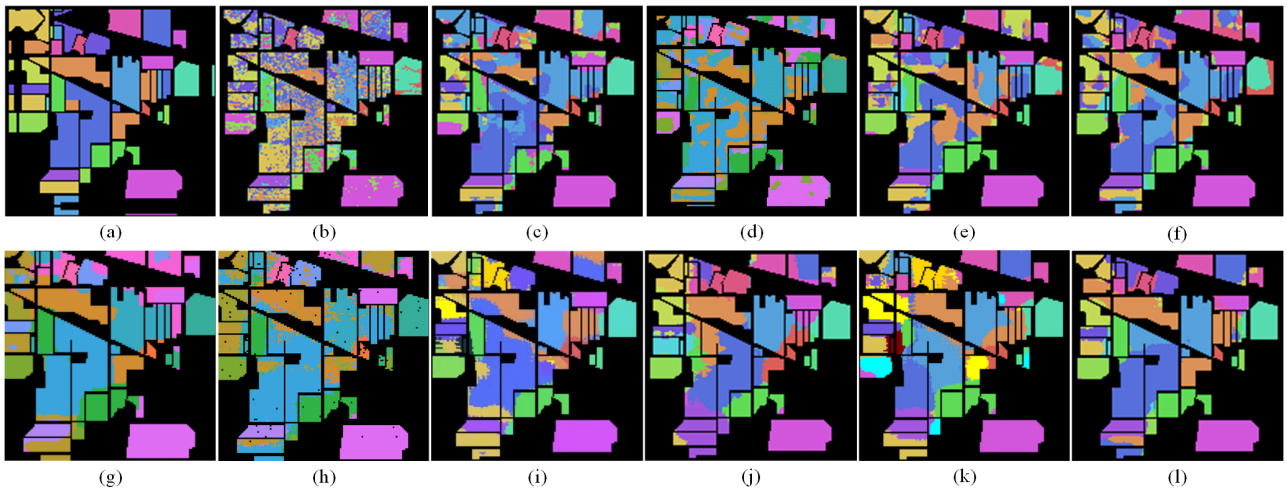

Fig. 7. Classification map of the Indian Pines dataset. (a) Ground truth. (b) SVM-RBF (46.65%). (c) 2DCNN (58.74%). (d) HiT (54.09%). (e) VIT (64.22%). (f) SSFTT (66.13%). (g) AMGCFN (75.48%). (h) DM-MRN (67.98%). (i) Res-CP (72.57%). (j) L2CSL (68.96%). (k) C-SS-MTr (72.47%). (l) PCL (76.71%).

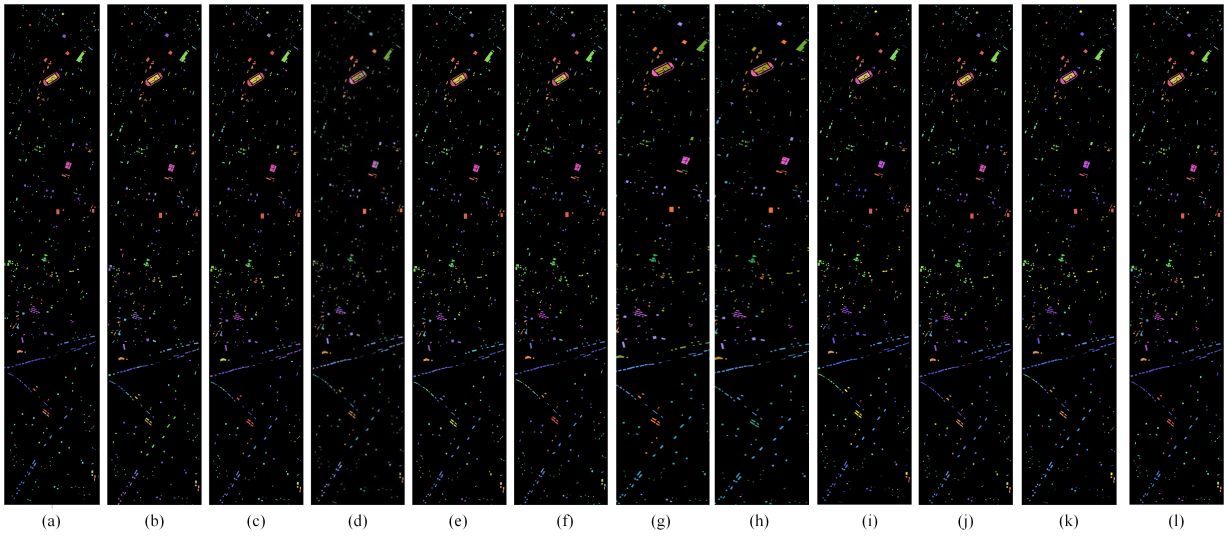

Fig. 8. Classification map of the Houston 2013 dataset. (a) Ground truth. (b) SVM-RBF (67.98%). (c) 2DCNN (69.58%). (d) HiT (71.86%). (e) VIT (70.09%). (f) SSFTT (72.61%). (g) AMGCFN (76.86%). (h) DM-MRN (76.56%). (i) Res-CP (80%). (j) L2CSL (78.02%). (k) C-SS-MTr (79.45%). (l) PCL (82.4%).

3) Strategies for Addressing False Negatives: False negatives present two critical issues in contrastive learning: discarding of semantic information and slow convergence. To address these issues, Huynh et al. [31] proposed the false negative cancellation method. This approach identifies false negatives within the same batch and treats them as positive samples to improve the model performance. To evaluate the effectiveness of the purified contrastive loss and compare its performance with that of the false negative cancellation, we conducted experiments on both strategies. The experimental results, as shown in Fig. 9, indicate that using the purified contrast loss consistently achieved higher OA across all three datasets compared with the use of the false negative cancellation approach. This observation suggests that the purified contrastive loss is a more effective strategy for reducing the impact of false negatives in the contrastive learning process. Typically, false negative samples in contrastive learning are not purely positive samples. By acknowledging this ambiguity and incorporating a weighting strategy based on similarity, the purified contrastive loss allows the model to focus more on learning from true positive samples while reducing the influence of false negatives, ultimately leading to improved performance.

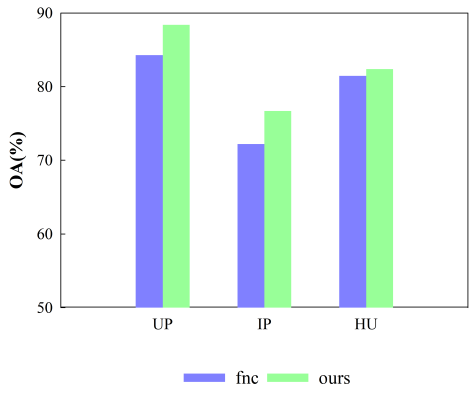

Fig. 9. Comparison of experimental results with false negative cancellation method for false negatives.

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TABLE VIII
FLOPs AND NUMBER OF PARAMETERS OF DIFFERENT METHODS ON HU DATA NUMBERS

Method	2D-CNN	HiT	ViT	SSFTT	AMGCFN	DM-MRN	Res-CP	L2CSL	C-SS-MTr	PCL
FLOPs(M)	13.8	739.8	15.7	11.4	56652.2	947	235.6	80.7	239.2	243.1
Param(M)	0.4	48.2	2.7	0.2	0.3	0.2	0.4	11.2	1.5	12.7

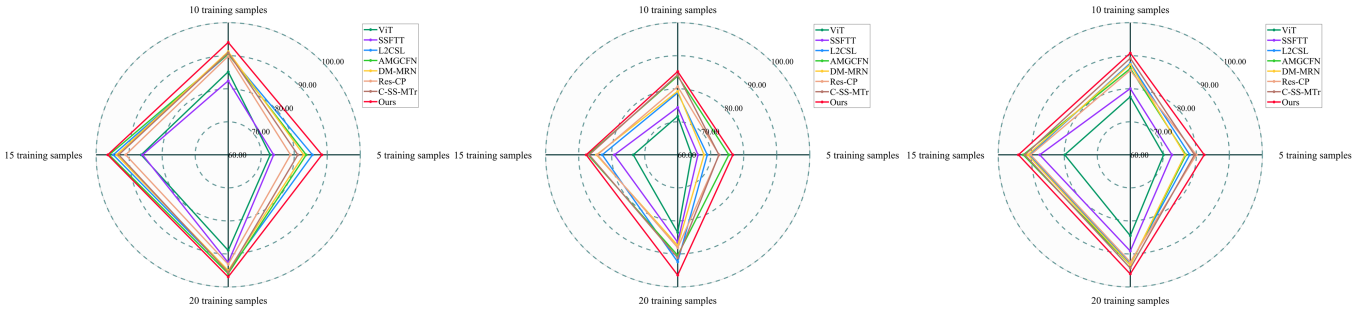

Fig. 10. Classification accuracy achieved with different numbers of training samples on three datasets. (a) UP. (b) IP. (c) HU.

TABLE IX
ABLATION STUDIES UNDER DIFFERENT LOSS FUNCTIONS ON THREE DATASETS

	UP		IP		HU	
With Weight	✓	×	✓	×	✓	×
OA(%)	88.39	85.77	76.71	74.96	82.40	79.23
AA(%)	91.65	86.87	87.03	84.10	84.90	81.56
$k\times100$	85.10	81.63	73.85	71.82	80.99	77.57

4) Complexity and Efficiency Analysis: To assess the efficiency of the proposed methods, the parameter numbers and floating-point operations (FLOPs) of all methods are detailed in Table VIII. It can be observed that the proposed method has a high FLOPs value, primarily because the proposed method incurs additional computation due to calculation of the similarity within hyperpixels. In addition, our encoder employs a hybrid two-branch structure, unavoidably leading to increased computational demands. Nevertheless, our method demonstrates a moderate level of performance in computational complexity.

D. Ablation Study

1) Ablation Study on the Purified Contrastive Loss: The main purpose of negative sample weighting loss is to alleviate the negative sample conflict problem to further improve the quality of contrastive training. To verify the effect of negative sample weighting loss in contrastive training, we replaced the loss function with InfoNCE loss function and conducted a series of comparison experiments on UP, IP, and HU datasets. We summarized the results under different loss functions, as shown in Table IX. It can be observed that the accuracy is improved after adding weight, especially on the HU is more significant with an improvement of nearly 3.17%. The results demonstrate that our purified contrastive loss can effectively reduce the interference of false negative samples by calculating the similarity between different samples, ultimately achieving optimal performance.

TABLE X
ABLATION STUDIES OF VARIOUS BRANCHES ON THREE DATASETS

Branch		Dataset		
DCNN	SSViT	UP	IP	HU
✓		84.12	75.68	80.54
	✓	82.66	72.29	79.68
✓	✓	88.39	76.71	82.40

TABLE XI
ABLATION STUDIES OF SSViT ON IP DATASET

Method	SpectralFormer	ViT	SSFTT	SSViT
OA(%)	86.53	95.66	97.47	98.58

2) Ablation Study on the DCNN-SSViT Encoder: The DCNN-SSViT encoder is designed to aggregate both global semantic features and local spatial features. To evaluate the effectiveness of the encoder for the HSI classification task under the InfoNCE loss (without weighting), we conducted an ablation study by separately removing the SSViT subnetwork and the DCNN subnetwork. The results presented in Table X show that the spatial–spectral information captured by the DCNN-SSViT encoder plays a more favorable role in the HSI classification task.

3) Ablation Study on the SSViT Network: SSViT is a global spatial–spectral feature extraction subnetwork specifically designed for the HSI classification task. To demonstrate the classification performance of SSViT, it is compared with state-of-the-art Transformer methods via a supervised learning classification task on the IP dataset. We randomly selected 10% of the total samples as the training set and the rest as the test set. The experimental results are shown in Table XI, and it can be observed that SSViT achieves the best OA score. With an OA of 98.58%, SSViT outperforms SSFTT by 1.11%, ViT by 2.92%, and surpasses SpectralFormer by over 12%.

E. Effects of Labeled Training Sample Numbers

To investigate the classification performance of our proposed model under different training samples, we randomly

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selected 10, 15, and 20 samples per class on the UP, IP, and HU datasets for comparison experiments, respectively. As shown in Fig. 10, the classification performance of all methods gradually improved with the increasing number of samples, which indicates that the number of labeled samples plays an essential role in training these deep learning models. Besides, the OA of our method is consistently higher than other methods under the same number of training sample. In conclusion, the experimental results show that our proposed method has better robustness and generalization ability.

IV. Conclusion

In this article, we proposed a purified contrastive self-supervised learning method for HSI classification, addressing the challenges associated with negative sample conflicts in contrastive training. Our main innovation is the introduction of a specialized purified contrastive loss designed to mitigate the impact of false negatives. In addition, we devised a complementary encoder, DCNN-SSViT, which integrates SSViT and DCNN branches, aimed at capturing both global abstract representations and local structural features. The SSViT branch employs the multiheaded attention mechanism to capture global spatial–spectral features, while the DCNN focuses on extracting structural information from local contexts. We conducted extensive comparison experiments and ablation studies on three datasets, and the results demonstrated the effectiveness of our proposed method.

Self-supervised learning is a hot research topic in recent years. Apart from the contrastive-based learning approach, another pixel-based reconstruction approach is also making a splash in the field of image processing. Understanding how these methods work and interact can provide a more comprehensive understanding of self-supervised learning techniques. In future work, we will explore the working mechanism of two approaches in more detail and pave the way for novel unified approaches that leverage the strengths of both approaches. In addition, we recognize the importance of comprehensively assessing the impact from various degradation, noise effects, or variabilities in the process of imaging on classification performance. A promising direction for our future research involves exploring the antidisturbance approach within the imaging process and developing classification models that against adversarial attacks.

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Lin Zhao (Member, IEEE) received the M.S. degree in computer application technology from Huazhong University of Science, Wuhan, China, in 2007, and the Ph.D. degree in control science and engineering from Central South University, Changsha, China, in 2017.

As an Associate Professor, he is conducting research programs in optical imaging and hyperspectral image processing. His research interests include image processing, pattern recognition, and modeling and optimal control of the complex industrial process.

Jia Li received the B.S. degree from Changsha University, Changsha, China, in 2021. He is currently pursuing the M.S. degree with Hunan Institute of Science and Technology, Yueyang, China.

His research interests include deep learning, pattern recognition, and hyperspectral image classification.

Wenqiang Luo received the M.S. degree from Hunan Institute of Science and Technology, Yueyang, China, in 2023.

His research interests include deep learning, computer vision, and remote sensing image processing.

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Er Ouyang received the M.S. degree from Hunan Institute of Science and Technology, Yueyang, China, in 2023. She is currently pursuing the Ph.D. degree with Shenzhen University, Shenzhen, China.

Her research interests include deep learning, pattern recognition and optical imaging, and object tracking.

Jianhui Wu received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 2000, and the M.S. and Ph.D. degrees in optical information engineering from Huazhong University of Science and Technology, Wuhan, China, in 2006 and 2009, respectively.

He was a Visiting Researcher at Computer Vision Group, University of Nevada, Reno, NV, USA, from August 2015 to August 2016, working on object detection and recognition. He is a Full Professor with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His main research interests include digital image processing and pattern recognition.

Guoyun Zhang (Member, IEEE) received the B.S. degree in industrial electrical and automation from Xiangtan University, Xiangtan, China, in 1993, and the M.S. and Ph.D. degrees in control theory and control engineering from Hunan University, Changsha, China, in 2000 and 2003, respectively.

In 1995, he joined the Hunan Institute of Science and Technology, Yueyang, China, where he is currently a Full Professor with the School of Information Science and Technology. He spent half a year (from January 2013 to June 2013) as a Visiting Scholar with George Fox University, Newberg, OR, USA. His research interests include image processing, pattern recognition, and digital signal processing application technology.

Wujin Li received the B.S. degree in software engineering and the Ph.D. degree in computer science and technology from Sichuan University, Chengdu, China, in 2007 and 2012, respectively.

His research interests include image denoising, image restoration, image enhancement, and hyperspectral image processing.

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